

Lower-bound witness
 OVE / checking
 Upper-bound decomposition
 Certificate infrastructure
 Held-out transfer
 Witness safety

Exoo 1989	■	□	□	□	□	□
Ge et al. 2022	■	■	□	□	□	□
Lehavi 2024	□	■	□	□	□	□
McKay–Radziszowski 1992	□	□	■	□	□	□
Angeltveit–McKay line	□	□	■	□	□	□
Gauthier–Brown 2024	□	□	□	■	□	□
This paper: H1 route	■	■	■	□	■	■

The matrix encodes the qualitative surfaces emphasized by each line of work. The intended novelty of the current paper is not new one-vertex extension itself, but the last two columns: cross-family transfer and witness-safe pruning. Those properties remain prospective.